SOFTWARE & APP MANUAL v2.0

BYB TELEMETRY



bybtelemetry.com

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SOFTWARE MANUAL



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HOW TO INSTALL?

Windows users

- Download the "**BYB Telemetry - WIN**" from BYB Telemetry repository
- Extract the installer from the .zip folder
- Click on the BYB Telemetry.exe file to begin the installation
- Follow the instruction to conclude the installation
- Launch the program

Macintosh users

- Download the "**BYB Telemetry - MAC**" from BYB Telemetry repository
- Extract the installer from the .zip folder
- Move the BYB Telemetry.dmg to the Application folder
- Follow the instruction to conclude the installation
- Launch the program

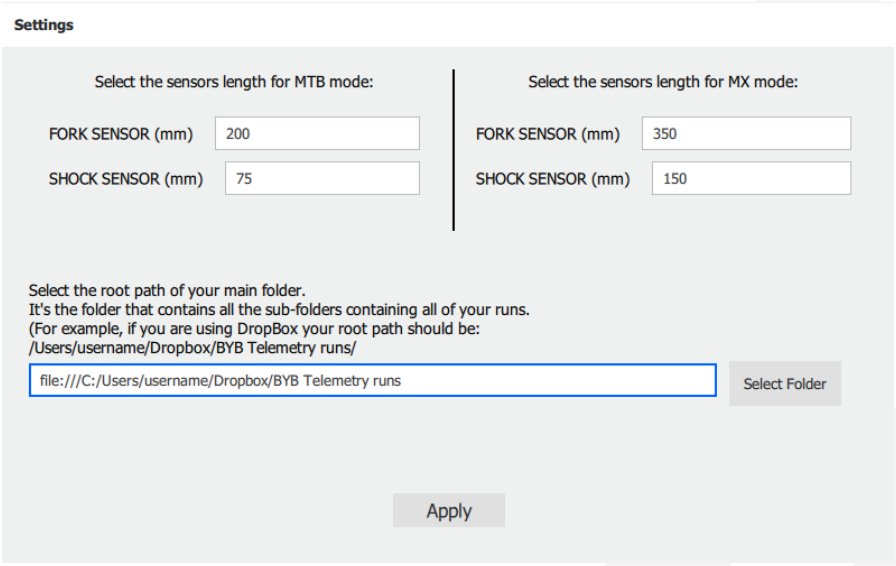
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PRELIMINARY CONFIGURATION

Set your sensors length

Click *settings* from the top menu bar. In order to visualize the data properly and make the auto-calibration effective is important to set the fork and shock sensor lengths. Default lengths are:

- MTB Fork 200mm
- MTB Shock 75mm
- MX Fork 350mm
- MX Shock 150mm



Root path

The root path defined by the user is automatically loaded at startup. Best practice is to set the root path to the main folder which contains all the .byb files.
If you are using the DropBox upload function from the smartphone, it's advisable to set the "BYB Telemetry runs" folder as your root folde. Click on the "Select folder" button to pick one.

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USER INTERFACE

Section of the software

- Menu bar
- File browser
- Control panel
- Tab selector
- GPS and altitude charts
- Tabs

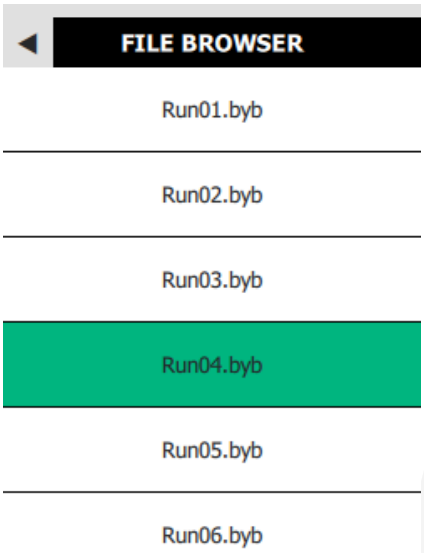


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FILE BROWSER

How to open a .byb file

The file browser will be set at startup accordingly to the root folder decided by the user.
To open a file is enough to navigate through the folders. *Double-click* on the folder name to advance, *single-click* on the top arrow to go back and *double-click* on the .byb file to open it in the software.



[TIPS]

To keep all the .byb runs tidy it's advisable to create a folder for every day of riding/testing. The folder name should contain the date and other information such as the place or the name of the rider.

Example: "*2021-02-15_Pila Bike Park*"

All of these folders should be placed inside your root folder.

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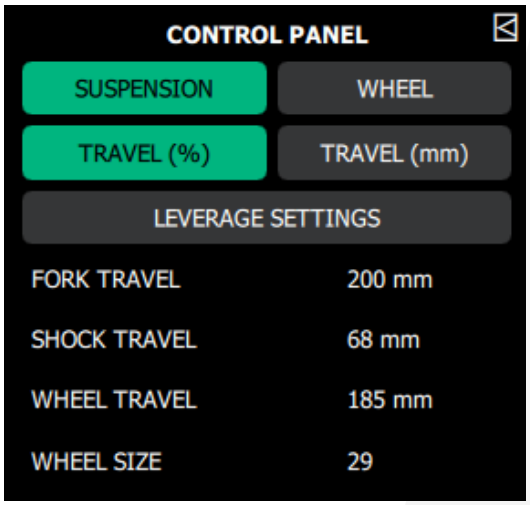
CONTROL PANEL

How to change display mode

The control panel shows the current display mode by highliting the respective buttons.
There are several choiches which can be freely selected:

- SUSPENSION - It's used to set the display mode to suspension travel
- WHEEL - It's used to set the display mode to front and rear wheel travel
- TRAVEL (%) - It's used to set the display mode to percentage
- TRAVEL (mm) - It's used to set the dispay mode to millimiters
- LEVERAGE - It's used to set the leverage ratio between the shock and the rear-wheel
- LAP MANAGER- It's used to obtain time splits and lap times

The software automatically updates all the charts and visualizations inside every other tab according with the selected parameters.



Like in the image above, the control panel contains the information that are automatically loaded from the .byb file too.

[TIPS]

If the values haven't been entered correctly, it's possible to modify them straigth from the "*Notes*" tab and click "*Save*". Once done, reload the run by double-click on the desider .byb file.

[CAUTION]

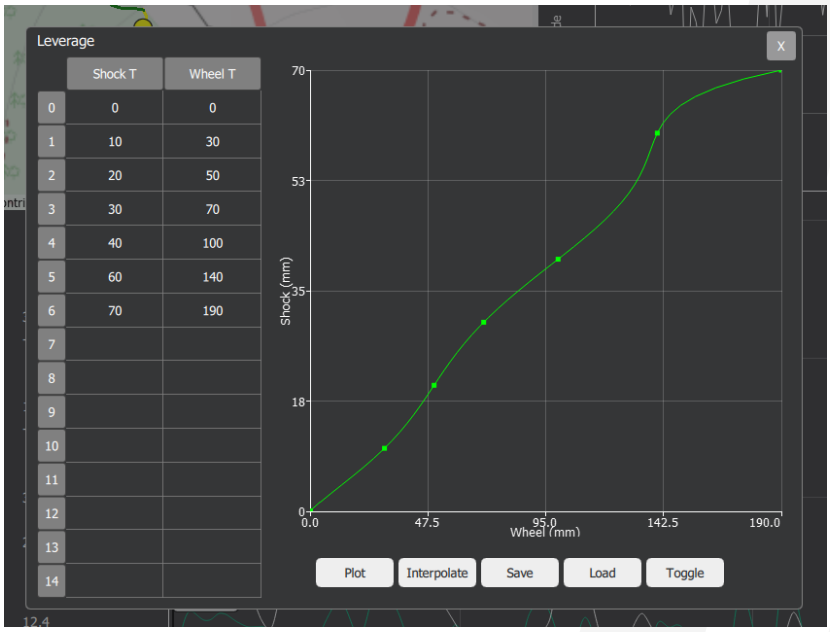
If "*Leverage settings*" button is not selected the wheel display mode for the rear wheel will be an approximation, using as reference the value of the bike rear travel.

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CONTROL PANEL

Leverage settings

Click on the "*Leverage settings*" button to open the leverage dialog.
Here you can set the parameters about the geometry of your frame. In particular, you can convert the shock travel into the rear-wheel travel.
The default leverage is linear and is automatically loaded from the data contained inside the notes.
The custom leverage curve is activated by filling the table.



How to plot the leverage curve

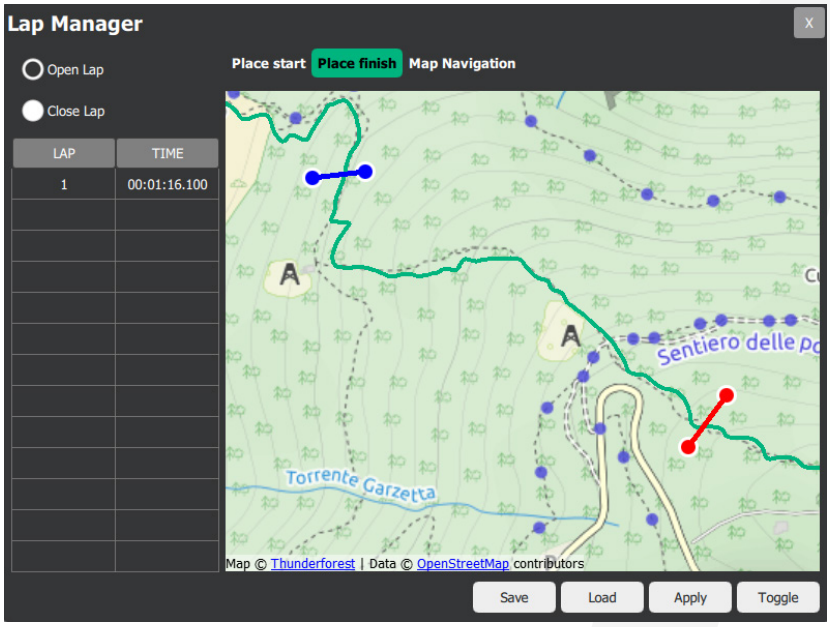
- Insert the values of shock travel and rear-wheel travel in the table (optimal amount is at least 6/7 points). As last point use max shock travel and max rear wheel travel
- Press the "*Plot*" button to display the points on the graph
- To generate the curve, press the "*Interpolate*" button. The software automatically calculates the conversion curve relative to your swingarm. Be sure that the displayed curve is passing through the points of the graph.
- Now you can select the "Wheel travel" option from the control panel
- You can save and load the leverage points of your bikes using the "*Save*" and "*Load*" buttons
- If needed, click the "Toggle" button to remove the custom interpolation

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CONTROL PANEL

Lap manager

Click on the "*Lap manager*" button to open the dialog.
Here you can set the "*start line*" and "*finish line*" if you are in open lap (usually used for downhill tracks), and you can set the "*start/finish line*" if you are in closed lap (usually used for motocross tracks).



How to visualize lap times and time splits

- Select open lap or close lap, according to the type of track you're riding.
- Draw on the map the "*start line*" and "*finish line*" or the "*start/finish" line*".
- Once done press "Apply". The stack view and suspension view will be automatically updated with the new range, from the selected start line to the selected finish line.
- Press "Save" to save the GPS coordinates in a file. This is usefull if you want to compare exactly the same time splits or lap times among diffrent runs.
- Press "Load" to load the GPS coordinate from a previously saved file.
- Press "Toggle" to exit from the Lap visualization mode.

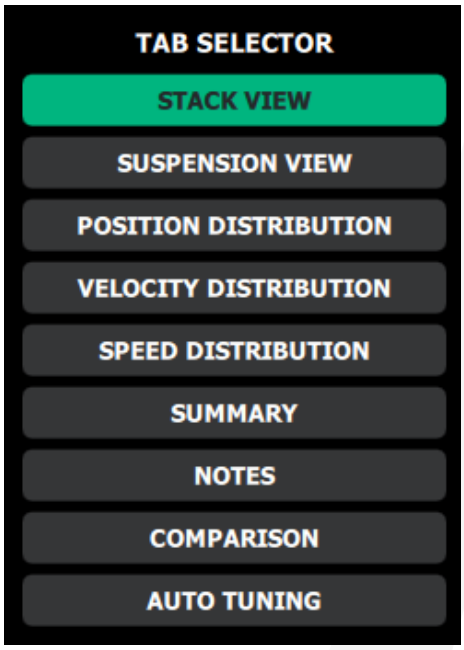
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TAB SELECTOR

Difference type of tabs

The software is characterized by several tabs. They are:

- STACK VIEW - It contains all the charts stacked togheter
- SUSPENSION VIEW - It containes detailed charts regarding every kind of data
- POSITION DISTRIBUTION - It contains the histograms about the bike's balance
- VELOCITY DISTRIBUTION - It contains the histograms of the suspension's velocities
- SPEED DISTRIBUTION - It contains the histograms of the rider's speed
- SUMMARY - It contains all the information and numbers about the opened run
- NOTES - It contains all the settings and notes regarding the current run
- COMPARISON - This advanced function is used to compare two runs.



[TIPS]

IIt's possible to hide the *tab selector* and the *control panel* in order to increase the charts' dimension by clicking on the upper left arrow.

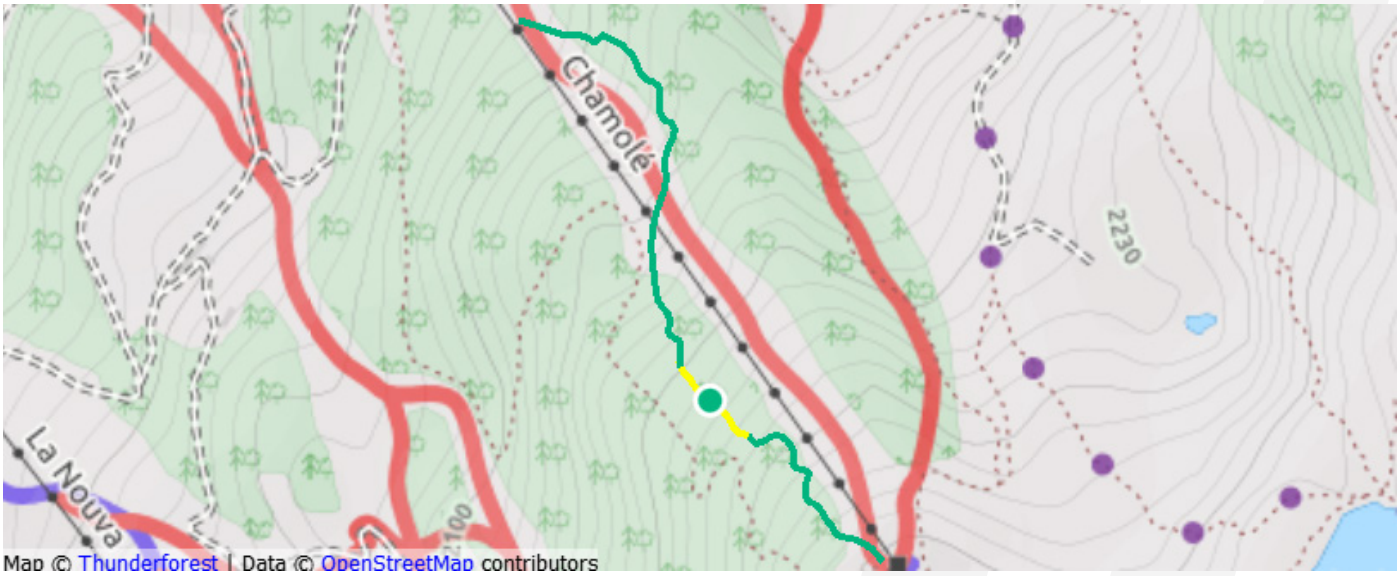
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GPS DATA

GPS map

The GPS map is automatically loaded when a .byb file is selected.
The route travelled by the rider is painted on the map, while the dot represent the current position of the rider which is animated by the cursor that scrolls the data.

The map can be zoomed and repositioned using the arrow and mouse wheel



Operations

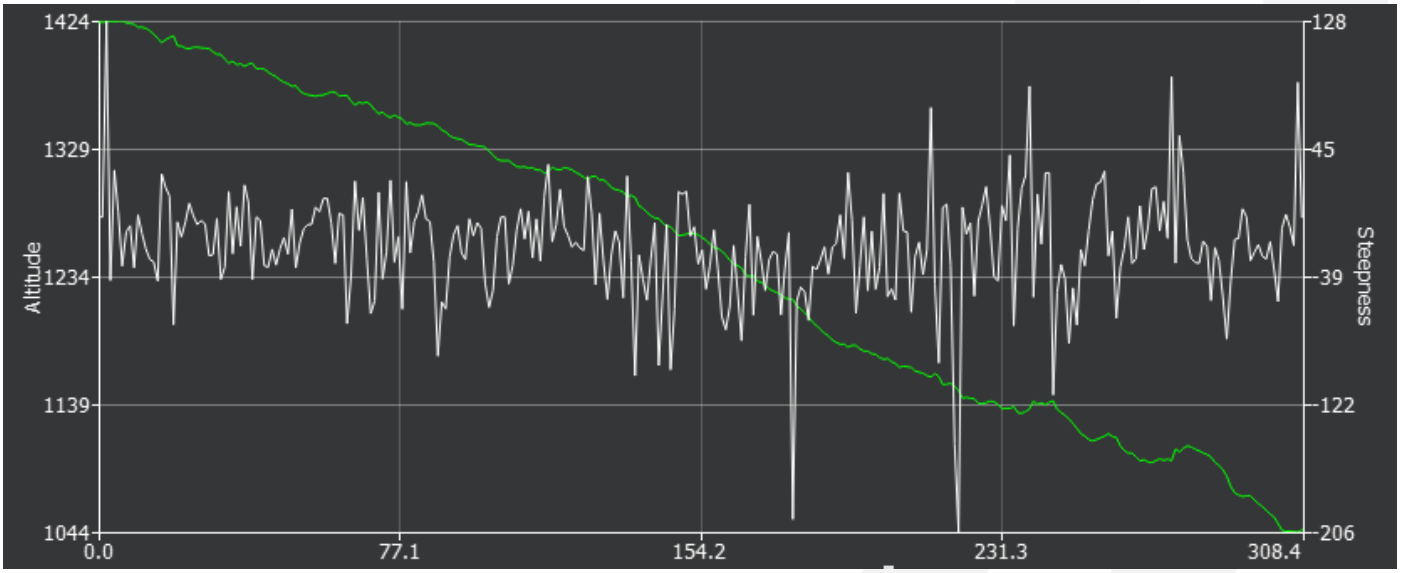
- Zoom the map
- Move the map

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GPS DATA

Altitude and steepnees chart

The altitude chart shows the vertical drop of the track.
The steepnees is calculated from the GPS data and is approximante. Negative values represent descends while positive values represent climbs.



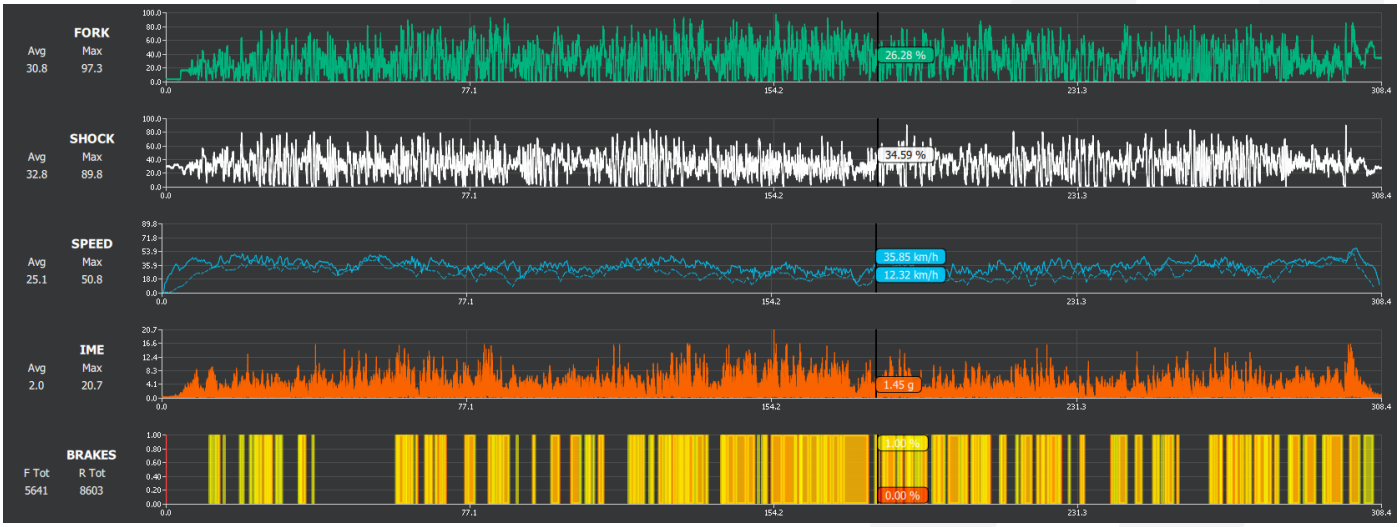
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TABS

Stack view

In this window, fork travel, shock travel, GPS Speed, sensor speed, Acceleromter (IMU) and brakes are shown. It's plotted also the comparison between the GPS and the sensor speed. The last one is characterized by a continue line and is much more responsive to sudden changes.
The IMU represents the shocks at the handlebar. This is the g-force and it's measured with an accelerometer placed inside the acquisition unit. The value is a multiple of the acceleration of gravity.
The brakes chart can be analog or digital, according to the sensors installed.



Operations

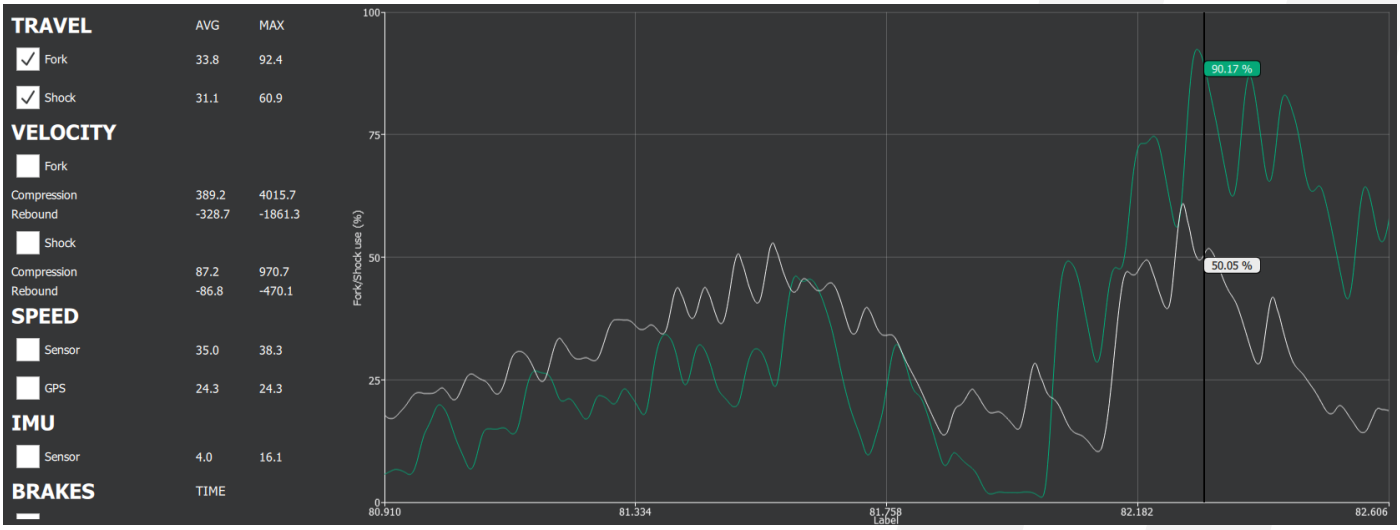
- Select any kind of display mode from the control panel
- Scroll the charts and move the GPS pointer
- Scroll the charts and read the current value from the pointer's labels
- Click, drag and release the mouse on any graph to zoom-in the interval. The corresponding part will be highlighted inside the GPS track too, while the average and the maximum values will be updated on the left
- In order to rescale the selection at the original size, double-click anywhere on the chart

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TABS

Suspension view

The suspension view is very important for a deep analysis. On the left side, you can select which data to show inside the chart.
It's included a banner which shows the time elapsed. It's mainly use to select a particular selection and to use the system as chronometer.



Operations

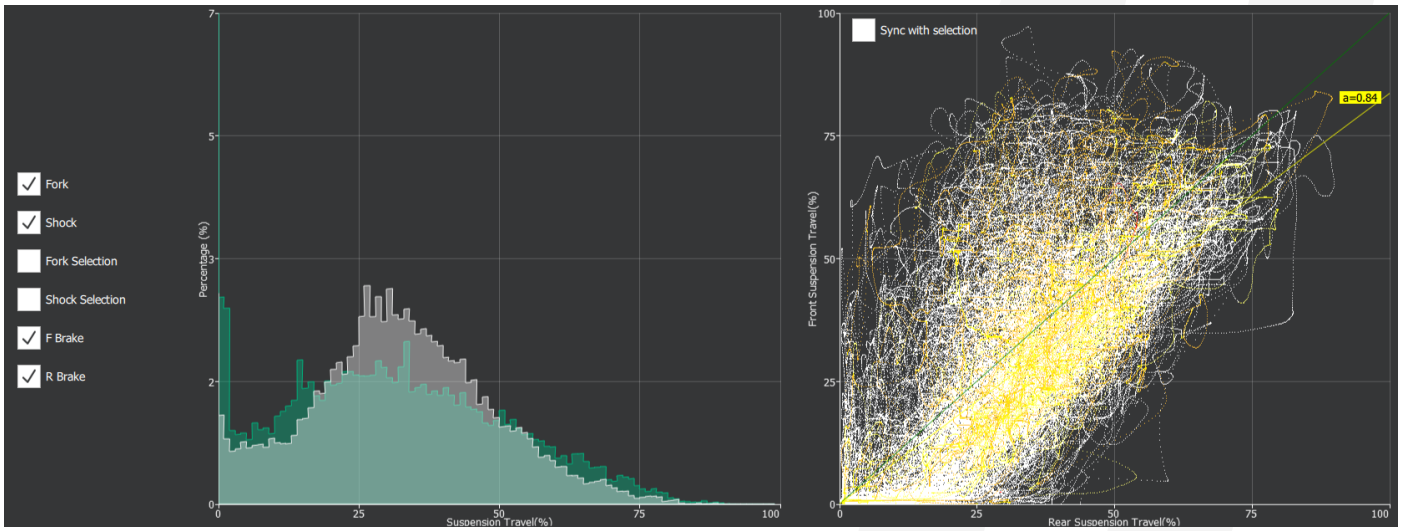
- Select any kind of display mode from the control panel
- Scroll the charts and move the GPS pointer
- Scroll the charts and read the current value from the labels
- Select which data to show and manage the overlay of the different lines
- Click, drag and release the mouse on any graph to zoom-in the interval. The corresponding part will be highlighted inside the GPS track too, while the average and the maximum values will be updated on the left
- The zoomed selection is shared an updated real-time between all the different tabs
- In order to rescale the selection at the original size, double-click anywhere on the chart

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TABS

Position distribution

The position distribution represents for how long (in percentage) your suspension stays at a certain value of travel. This is displayed through two histograms, one for the position of the fork and one for the position of the shock. It's a useful way to check the balance of the bike. To do this, it makes sense to plot the wheel position instead of the suspension position because wheel travel values differs a lot from shock travel values.
The chart on the right represents a dispersion of the points in which the coordinates are the front and rear travel at any time frame of the acquired run. The green line is a median which split the chart in two equal parts, while the yellow line is the best fit line for a perfectly balanced position between each wheel.



Operations

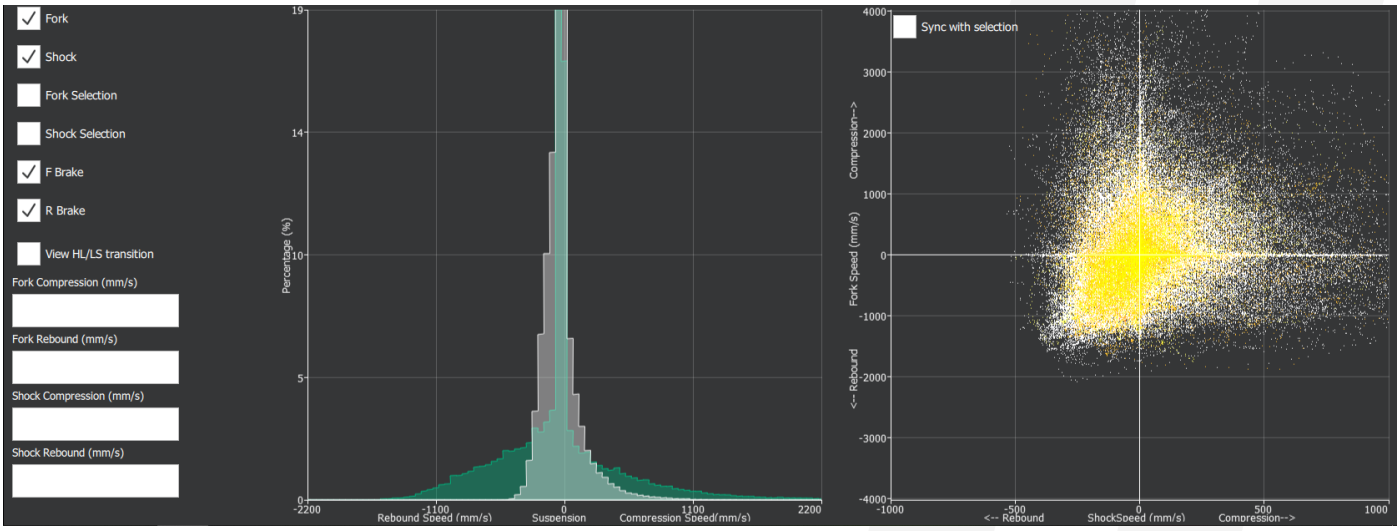
- Select any kind of display mode from the control panel
- Select which histogram to plot about the full run
- Select which histogram to plot about the current selection
- Select to sync the scatter plot with the current selection
- Select the overlay of brakes data

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TABS

Velocity distribution

The velocity distribution represents for how long (in percentage) your suspension stays at a certain value of speed. This is displayed through two histograms, one for the velocity of the fork and one for the velocity of the shock. It's a useful way to check the dyanamic balance of the bike. To do this, it makes sense to plot the wheel position instead of the suspension position because wheel speed values differs a lot from shock velocity values.
The chart on the right represents a dispersion of the points in which the coordinates are the front and rear velcoity at any time frame of the acquired run.



Operations

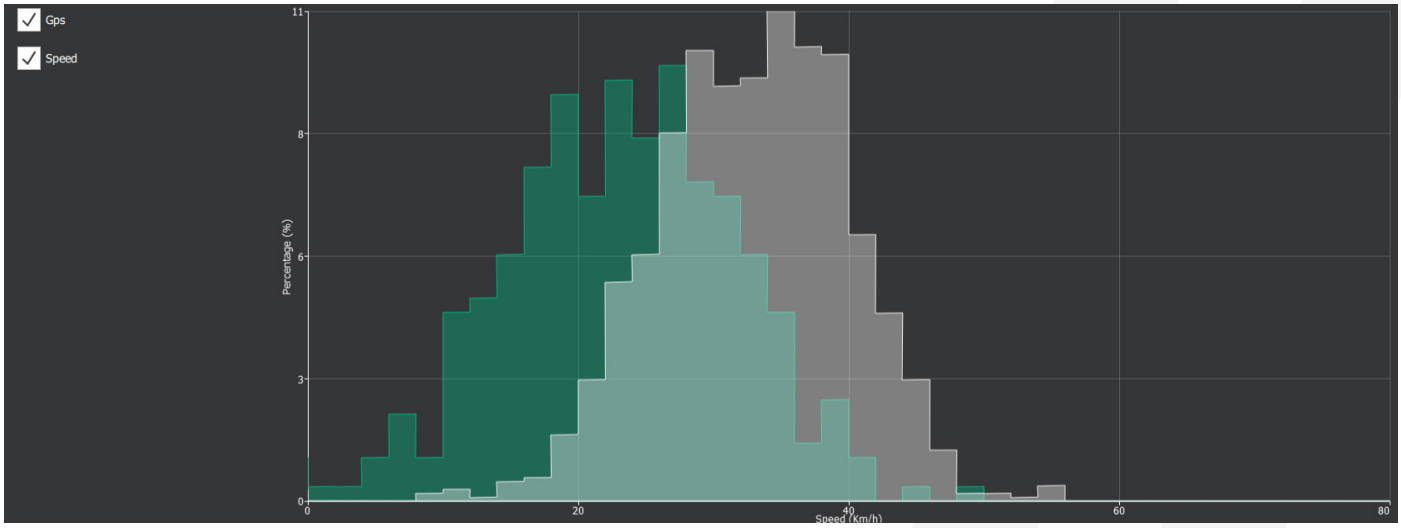
- Select any kind of display mode from the control panel
- Select which histogram to plot about the full run
- Select which histogram to plot about the current selection
- Select to sync the scatter plot with the current selection
- Select the overlay of brakes data

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TABS

Speed distribution

The speed distribution represents for how long (in percentage) a certain speed is sustained by the rider. It's normal that the GPS speed differs from the sensor speed because the first one is less accurate.



Operations

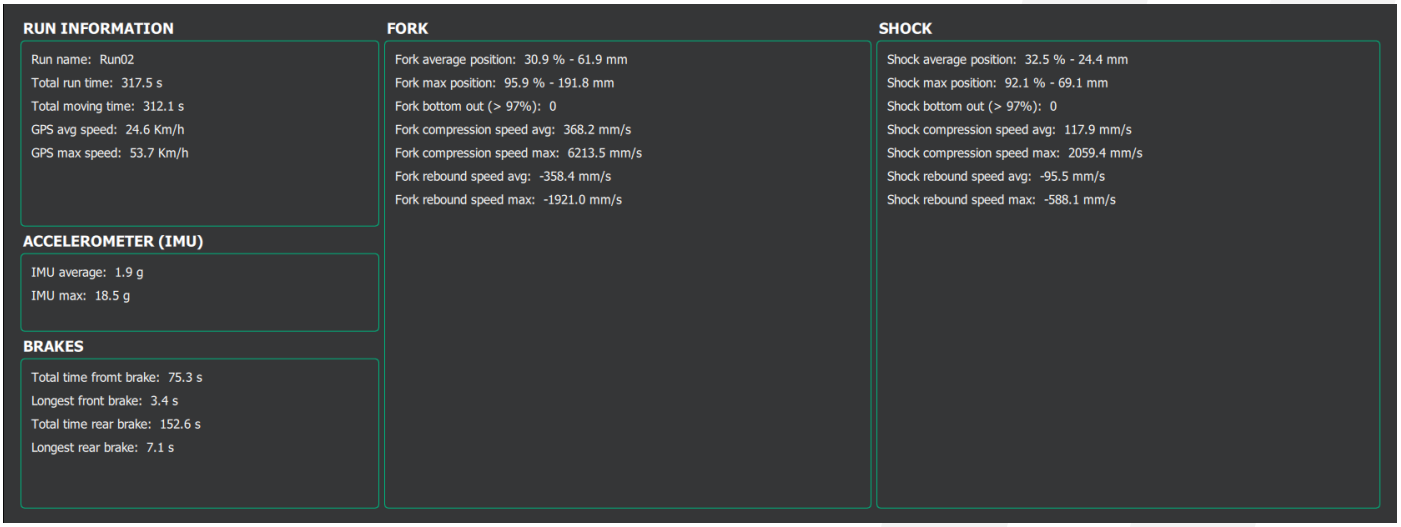
- Select to plot the histogram of the GPS speed
- Select to plot the histogram of the sensor speed

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TABS

Summary

The summary shows the relevant data about the current run.
There are several data available coming from the various sensors.



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TABS

Notes

The notes tab contains all the information, inputted by the rider/technician about the current setup. It's the best place to secure your settings and be able to consult them when needed.



Operations

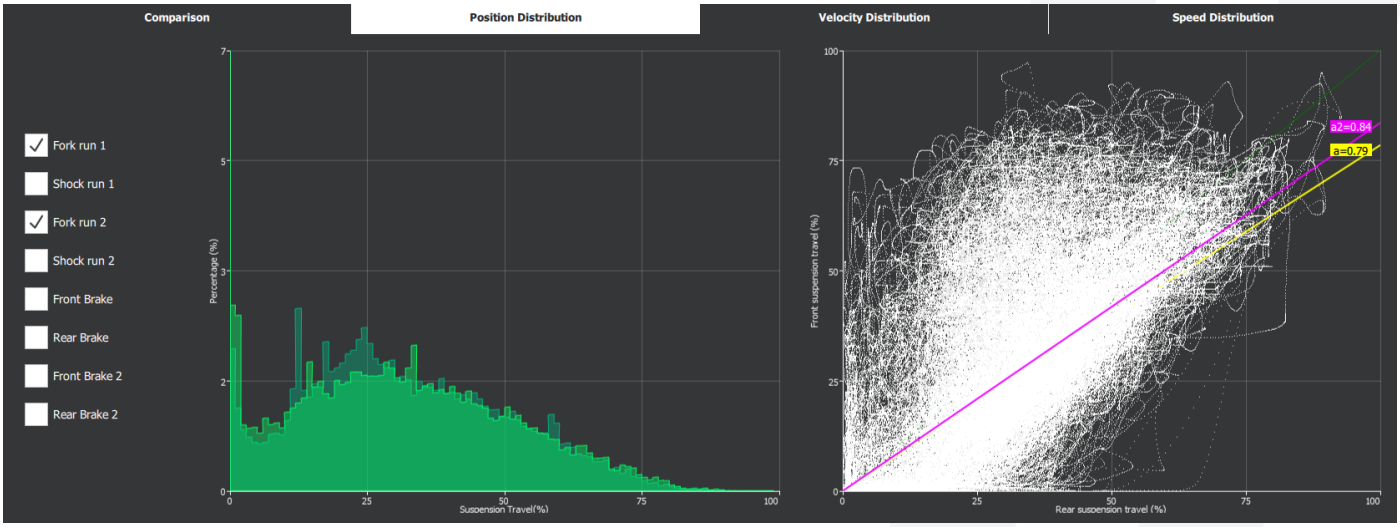
- Change the notes
- Save the changes

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TABS

Advanced comparison

This window allows you to compare and evaluate the gain/loss between two runs taken on the same track. The software automatically calculates the distance axis for both runs and displays the suspension travel in the top graph, while the gain or the loss of the first run compared with the second one is displayed in the bottom graph.
This tool is commonly used by racers and people who wants to improve their times and dig into the best set-up possible, taking into account all compromises of every section of the track.



Operations

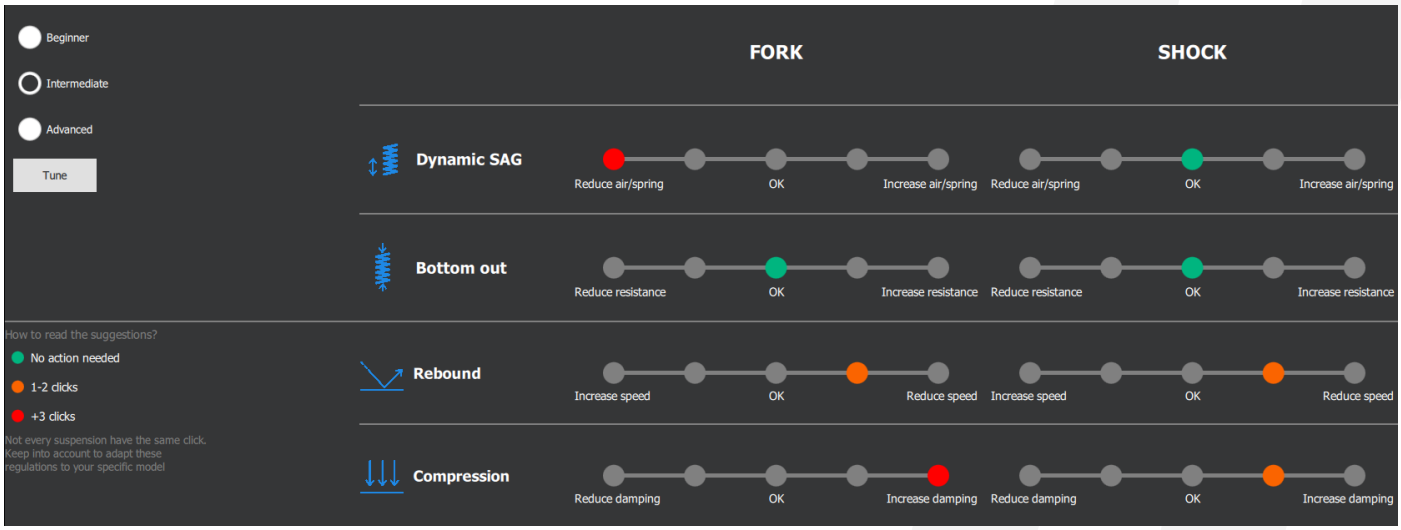
- Select any kind of display mode from the control panel
- Navigate between all the sub-tabs, such as the position, velocity and speed distributions tabs.
- Decide which vectors to plot

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TABS

Auto tuning (beta)

The auto tuning is a simple tool which uses a developed algorithm in order to find out if your suspension are working properly. The algorithm takes into account the level of the rider.
Depending on the result obtained is necessary to translate the suggestions on your bike. What to do is explained under each dot.



Operations

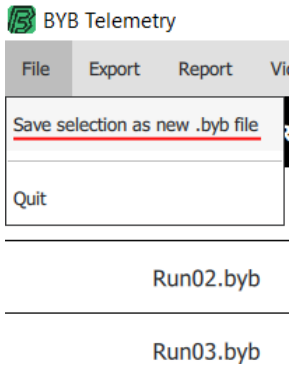
- Select the rider level
- Click the "Tune" button to view the results

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TRIM BYB RUNS

Save selection as new .byb file

This function is used to delete pauses or refine the start and stop of the runs.
To generate a new .byb trimmed file is necessary to select an interval inside the software from the stack view or the suspension view. Once the interval is set, click on "*File*" from the menu bar and select "*Save selection as new .byb*". This function will generate a new .byb file that reflects the selection and contains the notes and all source information.

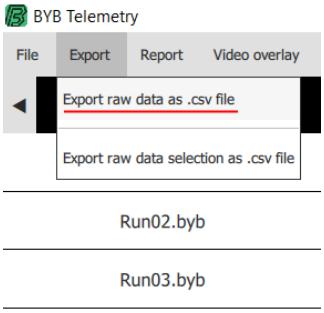


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EXPORT RAW DATA

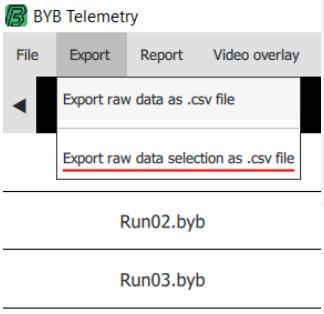
Export raw data as text

This function is used to export raw not-processed data to allow further custom analysis.
To generate the comma-seperated values file (.csv) is necessary to click on "*Export*" from the menu bar and select "*Export raw data as text*". This function will generate a new .csv file that contains only the full list of data points acquired divided into columns. Every column will include an header which represent the column title and the definition of the data type contained in that column.



Export raw data selection as text

This function is used to export raw not-processed data to allow further custom analysis. The difference between the previous function concerns the possibility of exporting data from a specific selection.
To generate the comma-seperated values file (.csv) is necessary to select an interval inside the software from the stack view or the suspension view. Once the interval is set, click on "*File*" from the menu bar and select "*Export raw data as text*". This function will generate a new .csv file that contains only the full list of data points acquired divided into columns. Every column will include an header which represent the column title and the definition of the data type contained in that column.

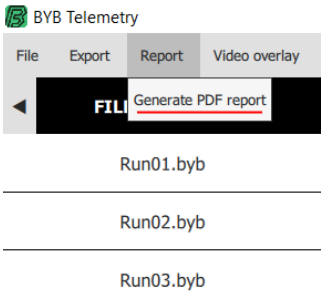


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EXPORT PDF REPORT

Export pdf report

This function is used to export a full .pdf of the entire run. To generate the pdf gile is necessary to click on "*Report*" from the menu bar and select "*Export report as .pdf*". This function will generate a new .pdf file that contains all the relevant information about the run.

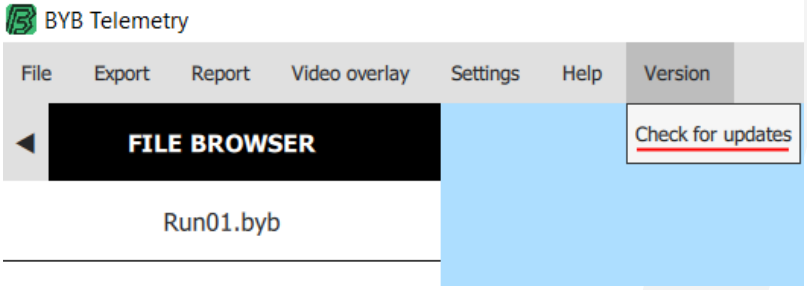


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CHECK FOR UPDATES

Check for updates

To check if your BYB Telemetry software is up to date, click on the “Check for updates” button from the main menu.
The software will perform a check at every startup too, and it will prompt a dialog if an update has been found.



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VIDEO OVERLAY

How to create video overlay using data and an action cam

Using this function is possible to create awesome video overlays. Basically, a video overlay is the animated overlay of data within a video.
The key point for an awesome video overlay is to synchronize the time frames.

It is necessary to proceed with the following steps:

- Start recording with the camera
- Start recording with the telemetry system
- Frame the suspension with the camera and perform a single squeeze. This will allow us to synchronize the timing of video and of the data easily
- Ride
- Stop the video and the acquisition once done



As in the image above, it's important to record with the camera a single compression of the fork. By doing this it will be easier to synchronize the time frames.

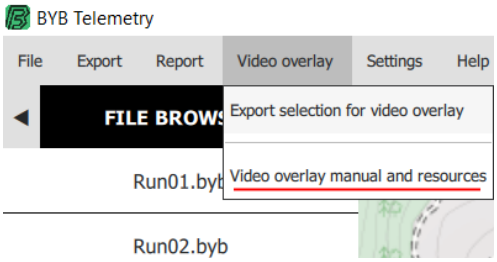
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VIDEO OVERLAY

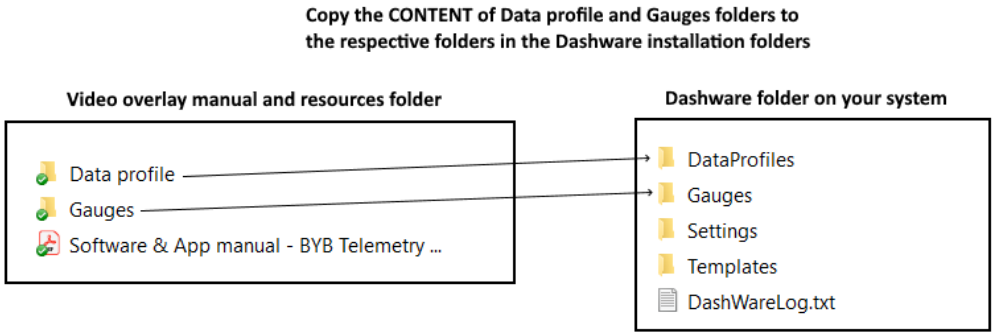
Post processing - One-time operations

The following operations need to be performed only the first time:

- Download the 3rd-party sofwware "Dashware" from: http://www.dashware.net/dashware-download/
- Click on the "Video overlay manual and resources" to download the packages needed



- Go inside the folder "Data profile" and copy the "BYB Telemetry v2.0.xml" file to ".../documents/Dashware/Data Profiles"
- Go inside the folder "Gauges" and copy all the files to ".../documents/Dashware/Gauges"

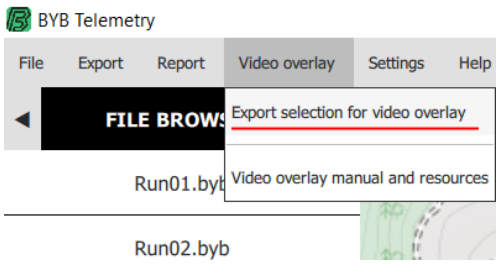


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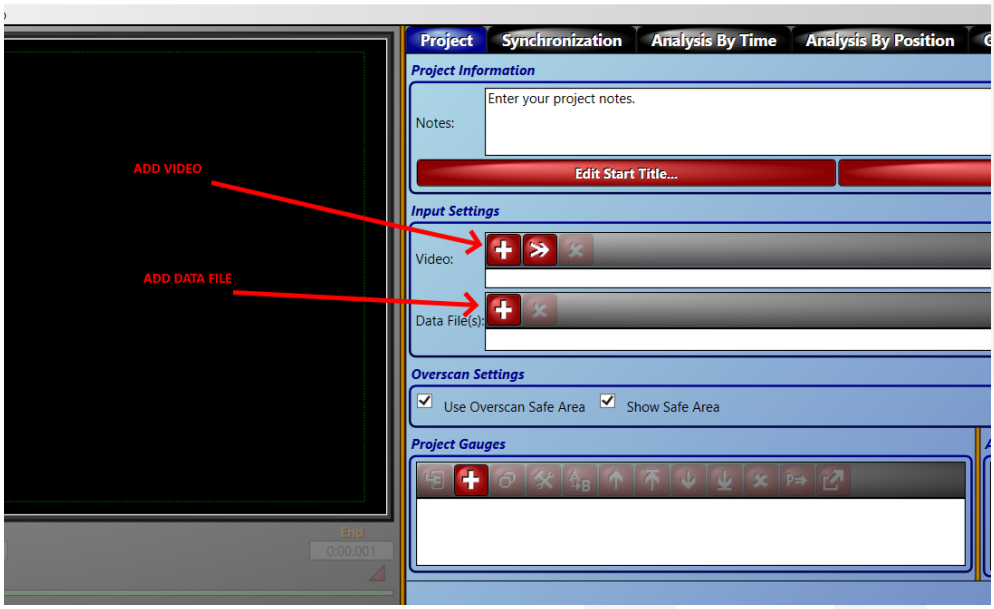
VIDEO OVERLAY

Post processing

• Open you run and click on the “Export selection for video overlay” on the main menu to export the current selection



• Open “Dashware” and create a new project
• Click “+” inside the “input settings” rectangle to add the recorded video
• Click “+” inside the “input settings” rectangle to ad the data file exported in the previous steps

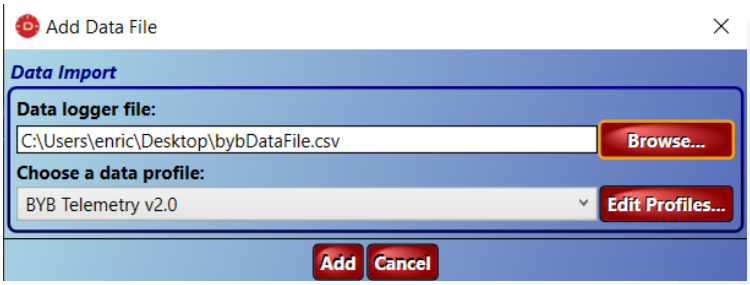


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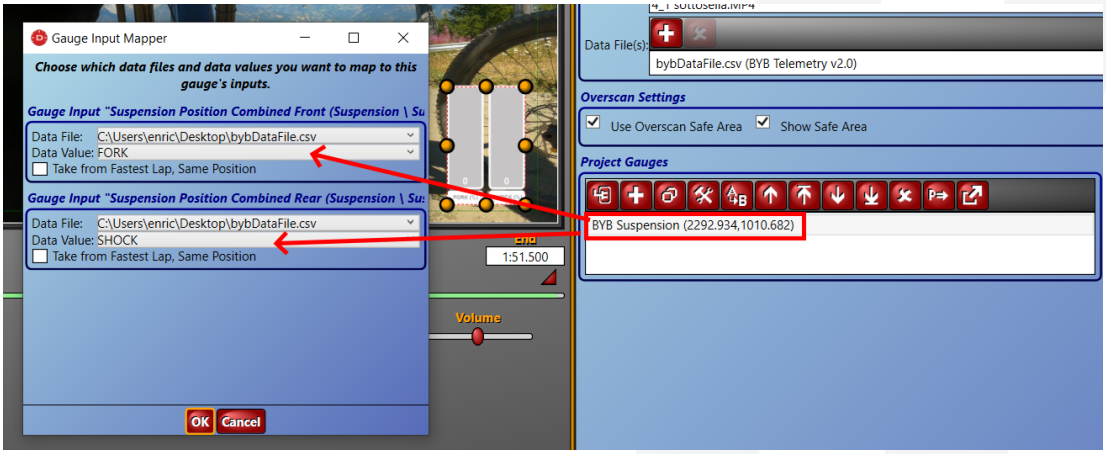
VIDEO OVERLAY

Post processing

- Browse the data file on your PC and select the data profile “BYB Telemetry v2.0”



- Drag and drop your “Gauges” on the screen from the “Gauges toolbox” panel
- Connect the gauges with the correct data source (Example: Suspension front > Fork)

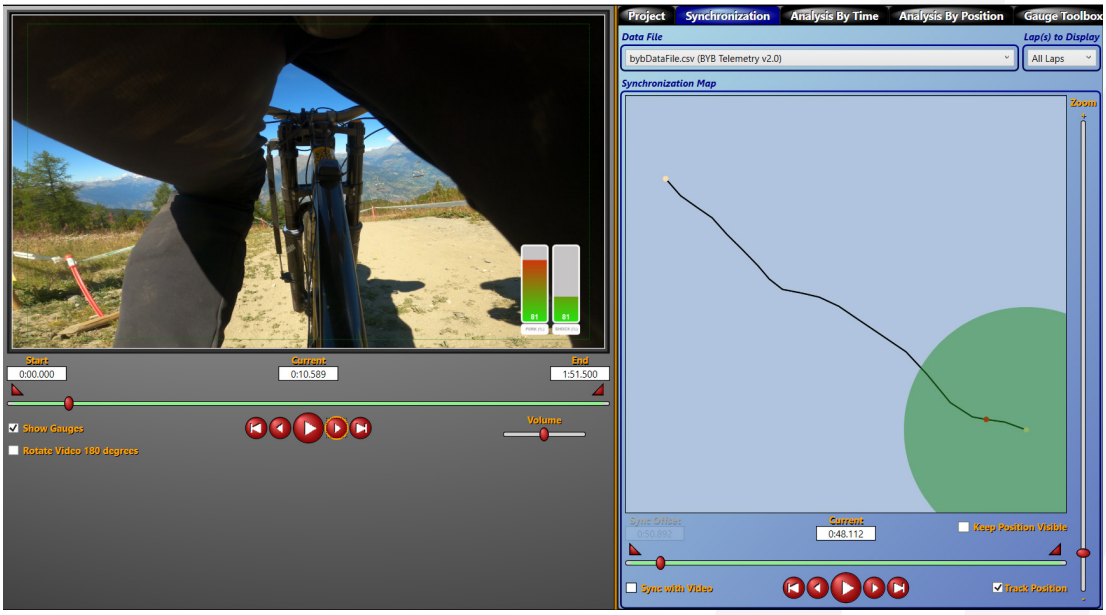


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VIDEO OVERLAY

Post processing

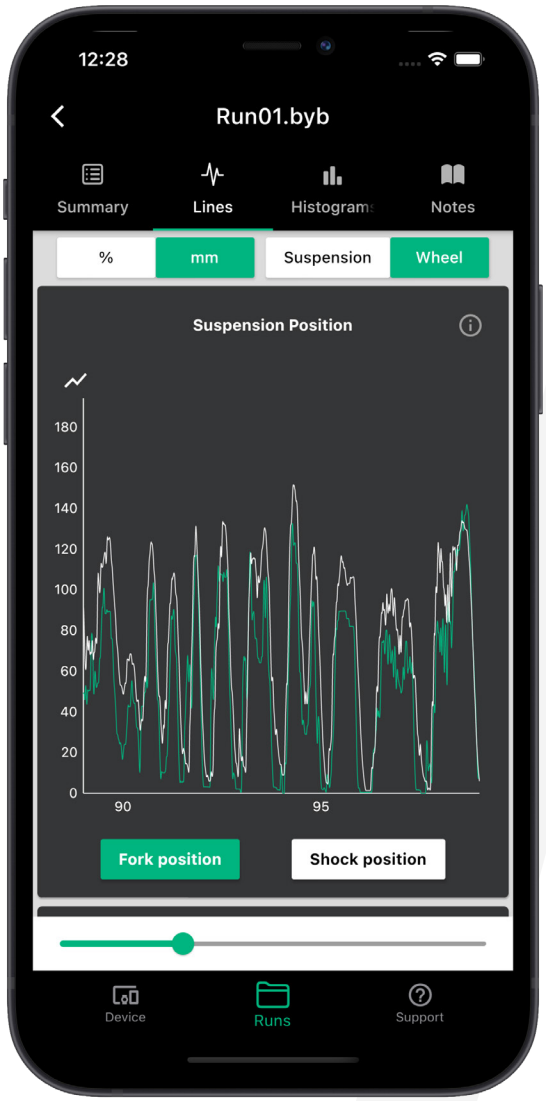
• Go into the “Synchronization” panel and match the time frame of video with the time frame of the data. Now it’s time to use the single compression we recorded in the video. Find the compression on the data and match it with the images on the video, exactly like in the image below.



• Once done click “Sync with video”
• Now you can export and render your video (File > Create video...)

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APP MANUAL



© BYB TELEMETRY

HOW TO INSTALL?

iOS users

The smartphone app is available for iOS smparthones and iPads.

- Download the "**BYB Telemetry**" app from the *Apple Store*

Download on the App Store

Android users

The smartphone app is available for Android smparthones and tablets.

- Download the "**BYB Telemetry**" app from the *Play Store*

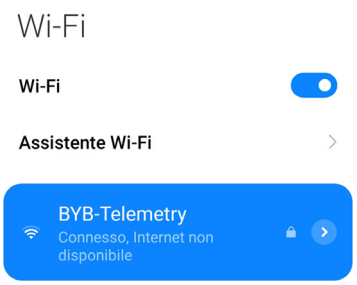
GET IT ON Google Play

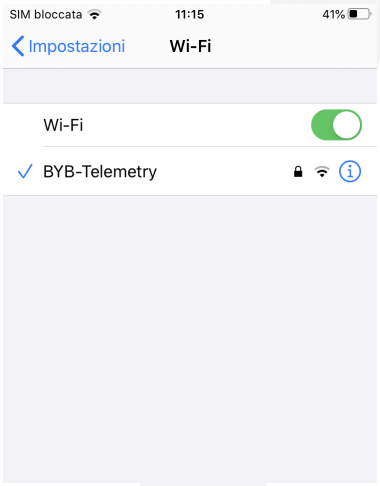
© BYB TELEMETRY

CONNECT THE ACQUISITION UNIT

Establish WiFi connection

To enable the connection between the smartphone and the acquisiton unit is necessary to:
- Enable the WiFi connection on the BYB Telemetry device by click the "Run button" for 3 seconds
- Once the run LED is blinking fast the device is discoverable by smaprthones and PC
- Open the WiFi settings of your smartphone and login to "**BYB-Telemetry**" networkd (if default)





Troubleshooting

Sometimes can happen that some smartphones reject the connection because "connection has not internet".
In this case is necessary to (according to the image on the rigth):
- Open "BYB-Telemetry" network WiFi settings
- Set the IP addres as "static"
- Type "192.168.1.1" as IP address



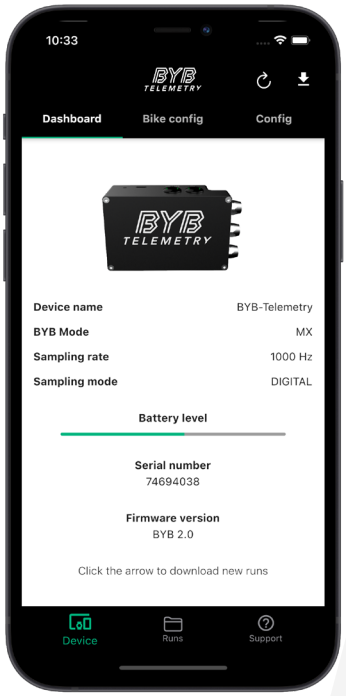
© BYB TELEMETRY

DEVICE PAGE

Dashboard

To check the current acquisition unit status is necessary to:

- Be sure the acquisition unit is connected to the smartphone
- Click on the “Device” tab and then click the “refresh” button in the upper rigth corner.



Here is possible to check the following configurations:

- Current device name
- Current “BYB Mode”
- Current “Sampling rate”
- Current “Sampling mode”
- Current battery level
- Device serial number
- Device firmware version

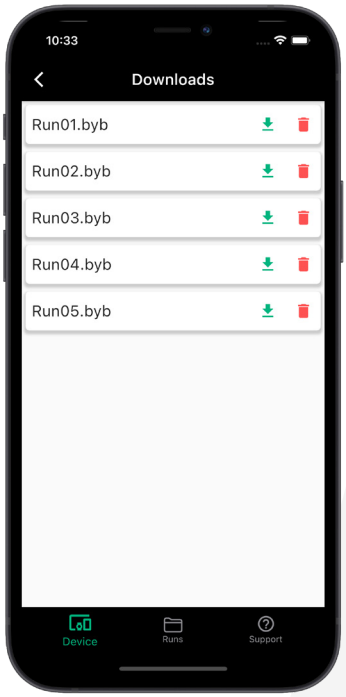
© BYB TELEMETRY

DEVICE PAGE

Download run

To download runs inside the device:

- After clicking on the downlaod button from the the "*Device page*" the following page is shown
- Click on the "*Download*" button to download a run from the device
- Click on the "*Delete*" button to delete a run from the device



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DEVICE PAGE

Bike config

To add personal information such as rider/bike settings or comments open the “Bike config” tab and fill the required fields.

- Fill the required fields
- Press “Save” once done
- Press “Delete”to empty all the fields

These data will be included as notes inside every new .byb file.



[TIPS]

ITo keep notes tidy it’s good practice to update the current “Bike config” every time you make changes to your bike setup. It’s advisable to update all the info before starting a new run too.

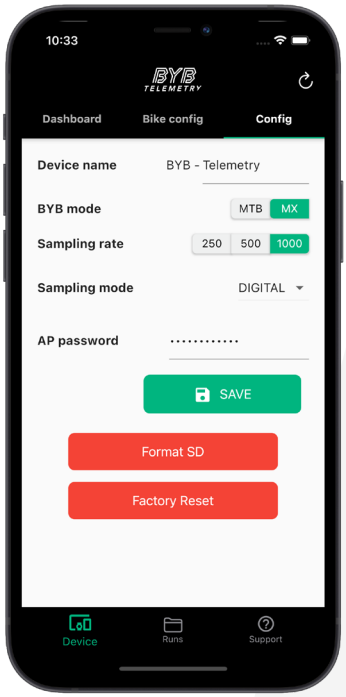
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DEVICE PAGE

Device config

To modify the acquistion unit configurations go the “Config” tab. From here it’s possible to:

- Modify the name of the acquisition unit (it’s equal to the network name too)
- Change the “BYB mode” between MTB and MX
- Change the “Sampling rate” between 250 Hz, 500 Hz, 1000 Hz (1000 Hz is reccomended)
- Change the “Sampling mode” between “DIGITAL” and “ANALOG” (brake sensors configurations)
- Change the WiFi network password
- Once done click “Save”
- To empty the memory of the device click “Format SD”
- To restore default configurations (including name and password) click “Factory reset”



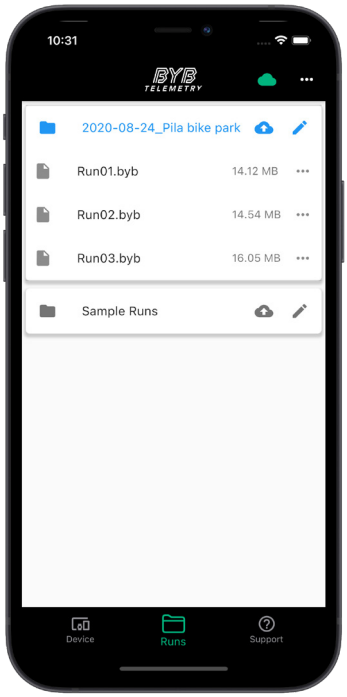
© BYB TELEMETRY

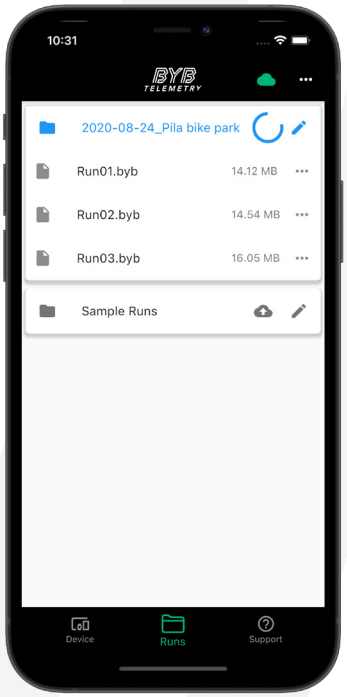
RUNS PAGE

Folders and Dropbox upload

Here are stored all the downloaded runs. Every run is downloaded inside a folder named with the date of the current day. The page offers the following possibilities:

- Rename the folders
- Rename the runs
- Delete a run
- Share a run outside the app
- Sign-in to Dropbox services (green cloud if Dropbox is already signed-in)
- Push runs to DropBox folder "BYB Telemetry runs" on your PC (click on the green cloud)
- Open a .byb file





[CAUTION]

After clicking on the run you want to open it may take several seconds before the run is loaded. This is due to the huge amount of data and calculations that must be performed.

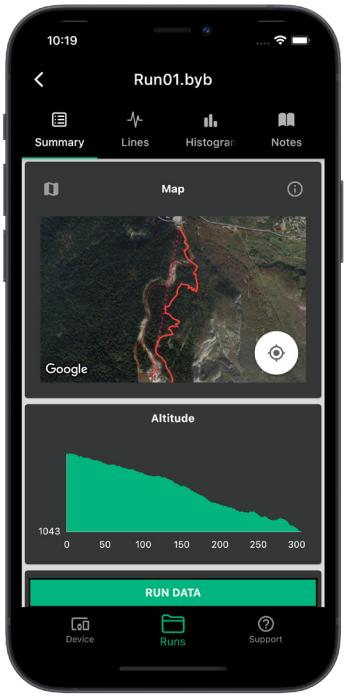
© BYB TELEMETRY

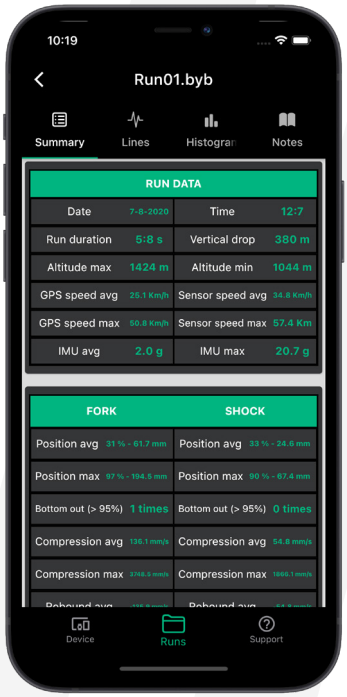
RUNS PAGE

Summary tab

The summary shows all the relevant data about the current run. The tab includes:

- The GPS map
- The altitude chart
- Run data
- Suspension data
- Brakes data





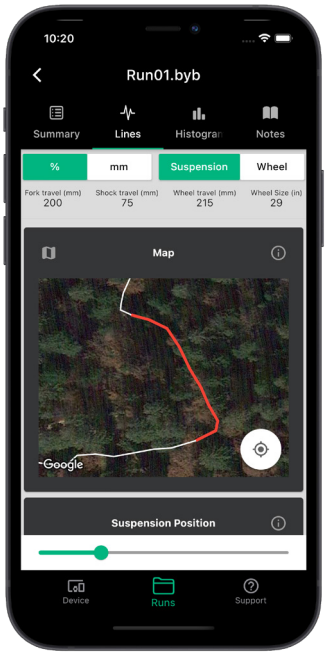
© BYB TELEMETRY

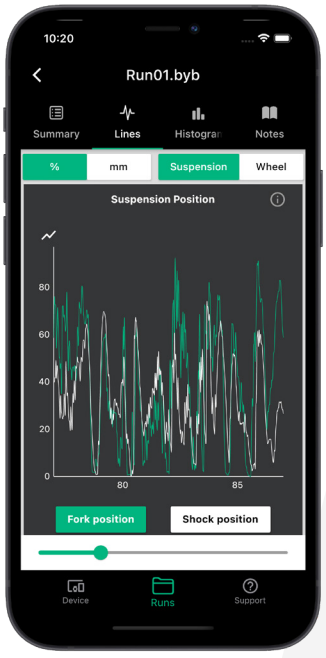
RUNS PAGE

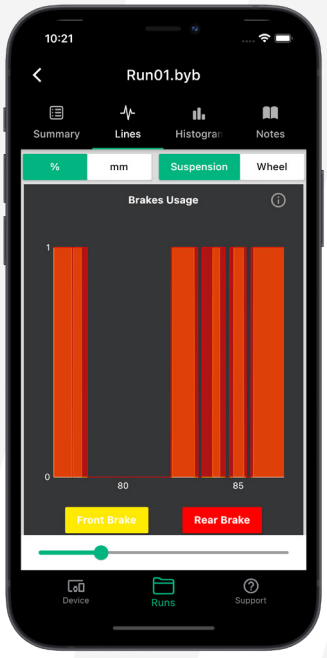
Lines tab

The lines tab shows the following charts:
- GPS map
- Suspension position
- Suspension velocity
- Accelerometer (shocks at the handlebar)
- Brakes
- Rider speed

It's possible to change the view selecting "*Suspension*", "*Wheel*", "*%*" or "*mm*".







The bottom slider is used to advance the time axis. All the charts are synchronized and controlled by the bottom slider.
To see the whole run it's necessary to bring the slider till the end. Representing all points can slow down the performance of the application. This is due to the huge amount of points

[CAUTION]

The charts may be different than the software because the graphic interface is lighter in order to match the smartphones capabilities.

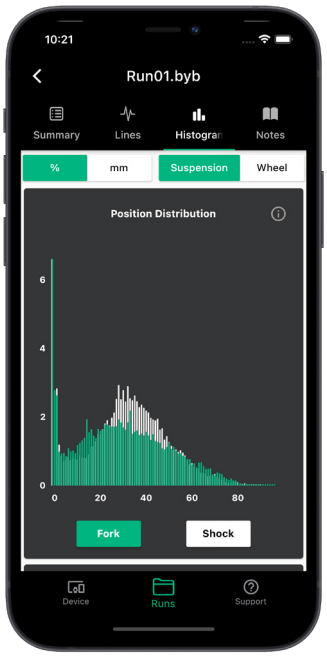
© BYB TELEMETRY

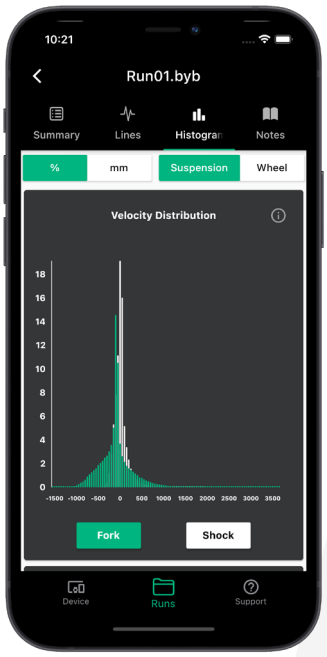
RUNS PAGE

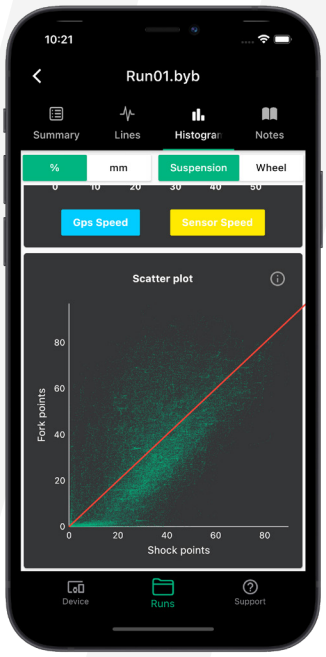
Histogram tab

The histogram tab shows the following charts:
- Position distribution
- Velocity distribution
- Speed distribution
- Scatter plot

It's possible to change the view selecting "*Suspension*", "*Wheel*", "*%*" or "*mm*".







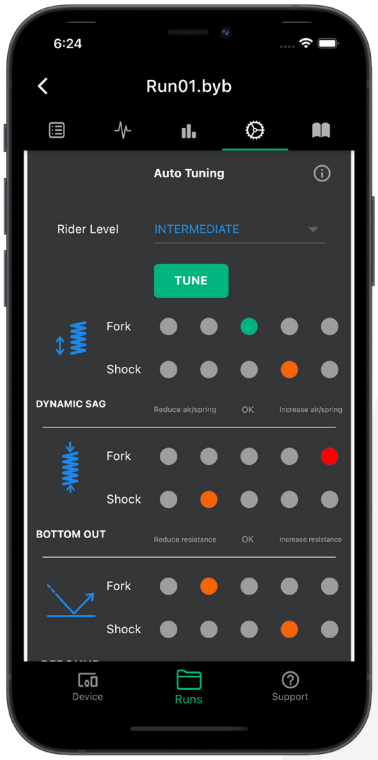
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RUNS PAGE

Automatic tuning tab

The auto tuning is a simple tool which uses a developed algorithm in order to find out if your suspension are working properly. The algorithm takes into account the level of the rider.
Depending on the result obtained is necessary to translate the suggestions on your bike. What to do is explained under each dot.

Select your rider level and click on the “Tune” button to see the results.



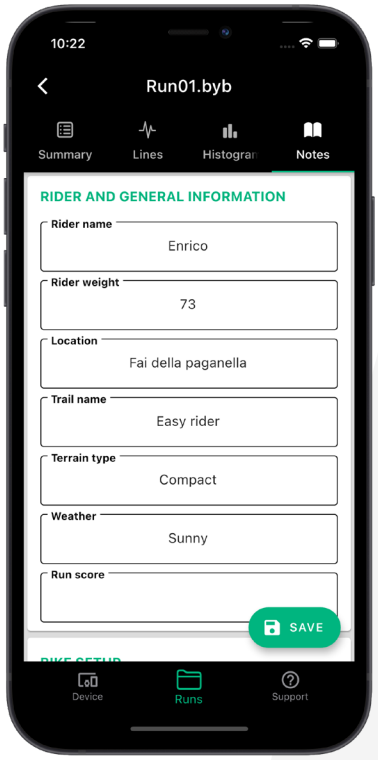
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SUPPORTS PAGE

Notes

The notes page contains all the configurations saved inside the .byb file.
It's possible to:

- Modify the already written fields
- Fill the empty fields
- Click "Save" to accept changes



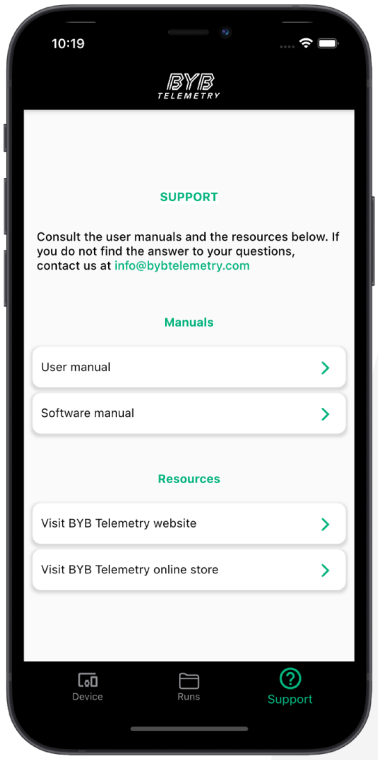
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SUPPORTS PAGE

Support

The support page contains some useful resource such as:

- Email address for any kind of support
- Consultable user manual
- Consultable software & app manual
- BYB Telemetry website
- BYB Telemetry online store



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